Emergency Lighting Battery – General data

1. Removal/Installation

A. Remove Emergency Lighting System Battery

CAUTION **BEFORE DISCONNECTING THE BATTERY, ENSURE THAT THE MASTER SWITCH IS OFF AND PULL COMP LTS CIRCUIT BREAKER. ANY SUBSTANTIAL LOAD ON THE BATTERY COULD CAUSE DAMAGE TO THE CONNECTOR PINS AND SOCKETS WHEN BATTERY IS BEING DISCONNECTED.**

Before removing the battery, refer to precautions in 24-00-00, Electrical Power – General Data.

1) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Remove the access panel in the rear of the baggage compartment floor. (Refer to 12-00-00, Servicing – General Data).

3) Disconnect the battery connector.

4) Remove the battery screws.

5) Remove the battery.

B. Install Emergency Lighting System Battery

1) Inspect the battery for damage. Cracks in metal or plastic containers are not permitted. Dents in the metal containers that impinge on the interior plastic container are not acceptable.

2) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Remove the access panel in the rear of the baggage compartment floor. (Refer to 12-00-00, Servicing – General Data).

4) Install the (4) screws to secure the battery.

5) Connect the battery connector.

6) Install the access panel in the rear baggage compartment floor.

7) Do operational test as per 33-50-00, Emergency Lighting – General Data, Para 2.A.

2. Adjustment/Test

A. Capacity Test

1) Stabilize the battery at 15°C or higher. The battery must be at this temperature for at least 24 hours.

2) Discharge the battery at the C1 rate on the label to the end point voltage (EPV) of 20 volts.

3) Record the time to the end point voltage (EPV).

4) The battery is acceptable for continued use if the ampere hour capacity (actual hours if discharge X ampere rate of discharge) is greater than 85% of the normal capacity (C1) shown on the battery label (i.e., 51 minutes or more).

5) If the battery passes capacity test, it can be charged as per and return to service.

6) If the battery fails the capacity test, condition battery as per Para 5.D.

7) After battery has been conditioned, repeat the capacity test.

Note If the battery gets very hot (external case temperature greater than 55°C (130°F) during charge, it should be replaced.

- If battery passes, refer to Para 2.A.5).
- If the battery fails, replace it.

3. Inspection/Check

A. Inspect Emergency Lighting Battery

A visual inspection in the aircraft should be made of the battery connector, receptacle and should be checked for contact failure or damage and the Allen head screws or nuts should be torqued, as per manufactures' data.

For further information refer to the manufacturers' data as follows:

1) Charge the Battery

a) Specials Tools:

- Advanced Power Products Beta D–50 Aircraft Battery Analyzer P/N 4126, or equal
- Advanced Power Products Alpha C–25 Battery Charger P/N 4142, or equal
- Advanced Power Products CA15–50 Charger/Analyzer P/N 4159, or equal

WARNING CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY THE BATTERY.

b) Depending on the type of charger available, charge the battery Constant Potential (CP). Charge at 2.35 volts/cell or 14.1 volts for 12 volt batteries, or 28.2 volts for 24 volt batteries until the charge current stabilizes for 1 hour.

2) Return to Service

a) Charge the battery as above. If the battery gets very hot (external case temperature greater than 55°C (130°F) during charging, it should be replaced.

4. Troubleshooting

Symptom	Probable Cause	Corrective Action
Low voltage/no voltage	Battery partially or fully discharged	Charge in accordance with Para 5.
	Battery fuse blown or circuit breaker tripped	Inspect and replace fuse Reset circuit breaker
Battery does not hold charge	Battery beyond serviceable life	Replace battery
Battery gets hot while recharging	Battery beyond serviceable life	Replace battery

5. Servicing Discharged Batteries

A. Recharging dual (parallel) batteries on the aircraft should be done individually.

B. Recharging a single battery on the aircraft:

1) Advanced Power Products Activator 282 Battery Charger, P/N 4105, or equal constant potential charger.

2) The Activator 282 is fully automatic. Connect to battery and initiate charging. If using a constant potential charger, follow instructions below.

CAUTION **ONLY CONSTANT POTENTIAL CHARGING MAY BE DONE ON THE AIRCRAFT. ENSURE THE CHARGING AREA IS WELL VENTILATED. DO NOT CONSTANT CURRENT CHARGE A BATTERY ON THE AIRCRAFT. THERE MAY BE A SERIOUS RISK OF INJURY TO PERSONNEL AND/OR DAMAGE TO THE AIRCRAFT AND AIRCRAFT SYSTEMS.**

WARNING **CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY THE BATTERY.**

- **SET THE CHARGER FOR 2.35 VOLTS PER CELL (14.1 VOLTS FOR A 12 VOLT BATTERY, 28.2 VOLTS FOR A 24 VOLT BATTERY).**
- **SET THE CURRENT LIMIT TO THE C1 RATE.**

Note If the current on external power unit is not adjustable, do not charge the battery in the aircraft.

- **INITIATE CHARGING AND CONTINUE TO CHARGE UNTIL THE CHARGE CURRENT STABILIZES FOR 1 HOUR.**

C. Uninstalled Recharging

1) Remove the battery from the aircraft.

2) Special Tools:

- Advanced Power Products Alpha C–25 Battery Charger, P/N 4142 or equivalent; Advanced Power Products CA15–50 Charger/Analyzer P/N 4159; or equal.

WARNING **CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY THE BATTERY.**

3) Depending on the type of charger available, charge the battery Constant Potential (CP). Charge at 14.1 volts for 12 volt batteries or 28.2 volts for 24 volt batteries until the charge current stabilizes for 1 hour.

Note Batteries that have been allowed to stand in a deeply discharged state may not accept a CP recharge. Refer to conditioning procedure in Manufacturer's Manual for handling those batteries.

D. Conditioning Procedure:

WARNING **THE BATTERY MUST BE REMOVED FROM THE AIRCRAFT PRIOR TO DOING A CONDITIONING CHARGE.**

1) Special Tools:

- Advanced Power Products Beta D–50 Aircraft Battery Analyzer P/N 4146 or 4159, or equal
- Advanced Power Products Alpha C–25 Battery Charger P/N 4142 or 4159, or equal

WARNING **CONTACT CONCORDE FOR DETERMINATION OF EQUAL TEST EQUIPMENT. SOME BRANDS OF BATTERY CHARGERS WILL DESTROY THE BATTERY.**

2) Procedure:

a) Discharge the battery at the C1 rate to an end point voltage of 9 volts for 12 volt batteries and 18 volts for 24 volt batteries.

b) Constant current charge at the C1/10 rate for 16 hours.

WARNING **THIS PROCEDURE MAY DAMAGE THE BATTERY IF DONE ON A REPETITIVE BASIS.**

c) Allow the battery to cool down for 8 hours.

d) Retest the battery following the capacity test, refer to Para 2.A.